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## Volume 02 — Advanced VAEs and Generative Adversarial Networks

Lectures L10-L18 · Weeks 4-6

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### L10: Conditional VAE (CVAE)

#### Learning objectives

- Extend VAEs to conditional generation  $p(x | c)$ .
- Derive the conditional ELBO.

#### Conditional generative model

Given condition  $c$  (class label, text caption, attribute vector):

$$p_{\theta}(x | c) = \int p_{\theta}(x | z, c) p(z | c) dz$$

The encoder becomes  $q_{\phi}(z | x, c)$  and the decoder  $p_{\theta}(x | z, c)$ .

#### Conditional ELBO

$$\mathcal{L}(\theta, \phi; x, c) = \mathbb{E}_{q_{\phi}(z|x,c)}[\log p_{\theta}(x | z, c)] - D_{\text{KL}}(q_{\phi}(z | x, c) \| p(z | c))$$

Typically  $p(z | c) = p(z) = \mathcal{N}(0, I)$ .

## Applications

- Class-conditional image generation (MNIST digits by label)
  - Text-to-image conditioning (caption as  $c$ )
  - Controlled attribute manipulation
- 

## L11: Beta-VAE and Disentangled Representations

### Learning objectives

- Understand disentanglement in latent spaces.
- Formulate the  $\beta$ -VAE objective.

### Disentanglement

A representation is **disentangled** if each latent dimension controls a single generative factor (e.g., rotation, color, scale) independently.

### $\beta$ -VAE objective

$$\mathcal{L}_{\beta\text{-VAE}} = \mathbb{E}_{q_\phi} [\log p_\theta(x | z)] - \beta \cdot D_{\text{KL}}(q_\phi(z | x) \| p(z))$$

With  $\beta > 1$ , stronger pressure on the KL term encourages **factorized** latent codes at the cost of reconstruction fidelity.

### Evaluation metrics

- **MIG** (Mutual Information Gap)
  - **SAP** (Separated Attribute Predictability)
  - **DCI** (Disentanglement, Completeness, Informativeness)
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## L12: Latent Space Interpolation and VAE Limitations

### Learning objectives

- Perform latent space interpolation and arithmetic.
- Identify fundamental VAE limitations.

### Latent interpolation

Given encodings  $z_1 = \mu_\phi(x_1)$  and  $z_2 = \mu_\phi(x_2)$ :

$$z(\alpha) = (1 - \alpha) z_1 + \alpha z_2, \quad \alpha \in [0, 1]$$

Decode  $\hat{x}(\alpha) = \mu_\theta(z(\alpha))$  for smooth morphing between inputs.

## Latent arithmetic

$$z_{\text{result}} = z_{\text{king}} - z_{\text{man}} + z_{\text{woman}}$$

## VAE limitations

| Limitation         | Cause                                           |
|--------------------|-------------------------------------------------|
| Blurry outputs     | Gaussian decoder assumption; MSE loss           |
| Posterior collapse | $\text{KL} \rightarrow 0$ ; decoder ignores $z$ |
| Weak samples       | ELBO is loose; $p_{\theta}(x)$ underestimated   |

These motivate adversarial training (GANs, next section).

## L13: Generative Adversarial Networks: Minimax Formulation

### Learning objectives

- Formulate GAN training as a two-player game.
- Derive the minimax objective and its practical variant.

### GAN architecture

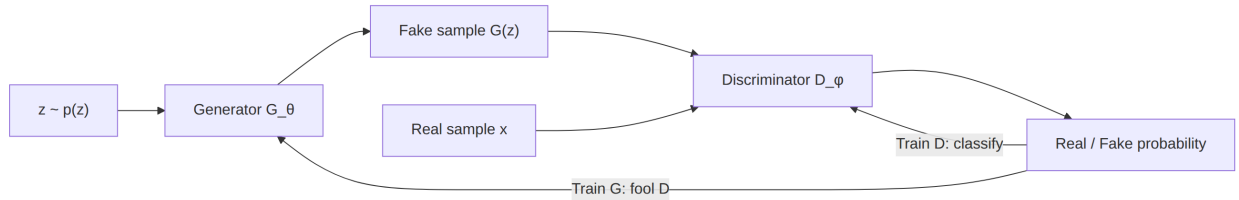


Figure 1: Architecture diagram

- **Generator**  $G_{\theta} : \mathcal{Z} \rightarrow \mathcal{X}$  maps noise to data.
- **Discriminator**  $D_{\phi} : \mathcal{X} \rightarrow [0, 1]$  classifies real vs. fake.

### Minimax objective

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

### Optimal discriminator

For fixed  $G$ , the optimal discriminator is:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)}$$

## Global optimum

At equilibrium,  $p_G = p_{\text{data}}$  and  $D^*(x) = \frac{1}{2}$ .

## Non-saturating generator loss

In practice, the generator minimizes:

$$\mathcal{L}_G = -\mathbb{E}_{z \sim p(z)} [\log D(G(z))]$$

This avoids vanishing gradients when  $D$  confidently rejects fakes.

## Key equation summary

$$\min_{\theta} \max_{\phi} \mathbb{E}_x [\log D_{\phi}(x)] + \mathbb{E}_z [\log(1 - D_{\phi}(G_{\theta}(z)))]$$

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# L14: GAN Training Dynamics and Mode Collapse

## Learning objectives

- Analyze GAN training instability.
- Describe mode collapse and mitigation strategies.

## Training challenges

| Problem             | Description                                     |
|---------------------|-------------------------------------------------|
| Mode collapse       | Generator produces limited variety              |
| Vanishing gradients | $D$ too strong; $G$ receives no useful signal   |
| Oscillation         | No stable Nash equilibrium                      |
| Non-convergence     | Alternating updates don't guarantee convergence |

## Mode collapse

The generator maps many  $z$  values to a few outputs, failing to cover the full data distribution. Detection: low diversity in generated samples; low inception score variance.

## Mitigation techniques

1. **Minibatch discrimination** — discriminator sees batch statistics.
2. **Unrolled GANs** — simulate future discriminator updates.
3. **Spectral normalization** — constrain discriminator Lipschitz constant.
4. **Wasserstein GAN (WGAN)** — replace JS divergence with Wasserstein distance:

$$\mathcal{L}_{\text{WGAN}} = \mathbb{E}_{x \sim p_{\text{data}}} [D(x)] - \mathbb{E}_{z \sim p(z)} [D(G(z))]$$

with weight clipping or gradient penalty on  $D$ .

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## L15: DCGAN Architecture and Design Principles

### Learning objectives

- Apply DCGAN architectural guidelines.
- Understand transposed convolutions for upsampling.

### DCGAN guidelines (Radford et al., 2016)

1. Replace pooling with strided convolutions (discriminator) and transposed convolutions (generator).
2. Use batch normalization in both  $G$  and  $D$ .
3. Remove fully connected hidden layers; use global average pooling.
4. Use ReLU in  $G$  (except output: tanh); LeakyReLU (0.2) in  $D$ .

### Generator structure

$z$  (100-d)  $\rightarrow$  FC  $\rightarrow$  4×4×512  $\rightarrow$  ConvT  $\rightarrow$  8×8×256  $\rightarrow \dots \rightarrow$  64×64×3

Each transposed convolution doubles spatial resolution while halving channels.

### Transposed convolution

For upsampling, transposed convolution (deconvolution) learns the upsampling kernel:

$$y_{i,j} = \sum_{m,n} W_{m,n} \cdot x_{\lfloor i/s \rfloor, \lfloor j/s \rfloor}$$


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## L16: Conditional GAN and Pix2Pix

### Learning objectives

- Condition GAN generation on labels or paired inputs.
- Describe the Pix2Pix image-to-image framework.

### Conditional GAN (cGAN)

$$\min_G \max_D \mathbb{E}_{x,y} [\log D(x,y)] + \mathbb{E}_{z,y} [\log(1 - D(G(z,y), y))]$$

Condition  $y$  is concatenated to input (channel-wise) or embedded and added.

## Pix2Pix

For paired image translation (e.g., sketch  $\rightarrow$  photo):

$$\mathcal{L} = \mathcal{L}_{\text{cGAN}} + \lambda \mathcal{L}_{\text{L1}}$$

where  $\mathcal{L}_{\text{L1}} = \|y - G(x)\|_1$  enforces pixel-level fidelity.

Uses a **U-Net generator** with skip connections and a **PatchGAN discriminator** that classifies local  $N \times N$  patches.

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## L17: CycleGAN and Unpaired Image Translation

### Learning objectives

- Translate between domains without paired training data.
- Explain cycle consistency loss.

### Problem setting

Domains  $X$  and  $Y$  with unpaired samples. Learn  $G : X \rightarrow Y$  and  $F : Y \rightarrow X$ .

### Cycle consistency

$$\mathcal{L}_{\text{cycle}} = \mathbb{E}_{x \sim p_X} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_Y} [\|G(F(y)) - y\|_1]$$

### Full objective

$$\mathcal{L} = \mathcal{L}_{\text{GAN}}(G, D_Y) + \mathcal{L}_{\text{GAN}}(F, D_X) + \lambda \mathcal{L}_{\text{cycle}}$$

### Applications

- Horse  $\leftrightarrow$  Zebra, summer  $\leftrightarrow$  winter, photo  $\leftrightarrow$  Monet painting
- 

## L18: StyleGAN and GAN Evaluation Metrics

### Learning objectives

- Understand StyleGAN's style-based generator.
- Apply FID and related evaluation metrics.

## StyleGAN innovations

1. **Mapping network** — transforms  $z$  to intermediate latent  $w$  in a less entangled space.
2. **Adaptive instance normalization (AdaIN)** — injects style at each layer:

$$\text{AdaIN}(x, y) = y_s \cdot \frac{x - \mu(x)}{\sigma(x)} + y_b$$

3. **Progressive growing** — train on low resolution first, then add layers.

## Evaluation metrics

### Fréchet Inception Distance (FID):

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

where  $(\mu_r, \Sigma_r)$  and  $(\mu_g, \Sigma_g)$  are mean and covariance of Inception features for real and generated images. Lower is better.

### Inception Score (IS):

$$\text{IS} = \exp(\mathbb{E}_x [D_{\text{KL}}(p(y | x) \| p(y))])$$

Higher IS indicates diverse, classifiable generated images.

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## Volume 02 summary

| Lecture | Core concept                     |
|---------|----------------------------------|
| L10     | Conditional VAE                  |
| L11     | $\beta$ -VAE, disentanglement    |
| L12     | Latent interpolation, VAE limits |
| L13     | GAN minimax objective            |
| L14     | Mode collapse, WGAN              |
| L15     | DCGAN architecture               |
| L16     | cGAN, Pix2Pix                    |
| L17     | CycleGAN                         |
| L18     | StyleGAN, FID/IS                 |

**Previous:** [Volume 01](#) · **Next:** [Volume 03 — Diffusion and Transformers](#)